

RSA (Decryption)

Proof of correctness

$$m = c^d \bmod n$$

$$\underbrace{m = c^d \bmod p \quad \text{and} \quad m = c^d \bmod q}_{\text{by CRT}} \quad \begin{aligned} m &= c^d \bmod pq \\ m &= c^d \bmod n \end{aligned}$$

$$\begin{aligned} m &= c^d \bmod p \\ &= m^{ed} \bmod p \end{aligned}$$

$$\begin{aligned} ed &= 1 \bmod \phi(n) \\ ed &= 1 + k\phi(n) \\ &= k(p-1)(q-1) + 1 \end{aligned}$$

$$\begin{aligned} m^{ed} &= m \cdot m^{k(p-1)(q-1)} \bmod p \\ &= m \cdot \left[\frac{m^{(p-1)}}{p} \right]^{k(q-1)} \bmod p \\ &= m \bmod p \end{aligned}$$

Similarly we can do for q .

RSA - OAEP — encry

RSA - PKCS# V.1.5 — encry

PSS — signature

Chosen ciphertext attack

$$C = m^e \bmod n$$

to

Attacker

$$x \in \mathbb{Z}_n^*$$

$$Y = C * x^e \bmod n$$

→ send this y to Bob

Decryption

$$y^d \bmod n = Z$$

Hacker can easily find m .

$$Z = y^d \bmod n = (C * x^e)^d \bmod n$$

$$= C^d * x^{ed} \bmod n$$

$$= \underline{(C^d * x) \bmod n} \quad ($$

$$\underline{ed = k * \phi(n) + 1}$$

$$m * x \bmod n = Z$$

$$m = Z^{-1} * x^{-1} \bmod n$$

Short message attack

hello team

permutation
digits of length 9

Primality testing

1. Choose an odd integer p .
2. Test if it is a prime number or not.

Fermat's little theorem

let p be a prime and $a \perp p$ ($\gcd(a, p) = 1$)

Then $a^{p-1} \equiv 1 \pmod{p}$ — necessary condⁿ for p to be prime

if not satisfy p is not prime.

Fermat's primality test

select a ~~prime~~ p .

choose 'a' $\perp p$ randomly

compute $\gcd(a, p)$

If $\gcd(a, p) > 1 \rightarrow p$ is not a prime

return p is composite number

else

$a \perp p$

compute

$$y = a^{p-1} \pmod{p}$$

if ~~$y = 1 \pmod{p}$~~ $y \neq 1 \pmod{p}$

then return p is composite

else

1. p — prime (probabilistic sense)

↓

prime base a

example

$$p = 49$$

$$a = 19$$

EEA

$$\boxed{\text{GCD}(19, 49) = 1}$$

$$y \equiv 19^{48} \pmod{49}$$

$$\equiv 1 \pmod{49}$$

$$\boxed{49 \text{ is a prime}}$$

(probable)

49 is a prime base 19

$$p = 49$$

$$a = 5$$

$$\text{GCD}(5, 49) = 1$$

$$y = 5^{48} \pmod{49}$$

$$48 = \underline{00110000}$$

$$y \not\equiv 1 \pmod{49}$$

49 is not prime

$$48 = \underline{00} \underline{11} \underline{0000}$$

$$5^{48} \pmod{49}$$

$$(1) \quad z = 1$$

$$z = 1 \pmod{49}$$

$$(2) \quad z = 1 \pmod{49}$$

$$(3) \quad z = 1 \pmod{49}$$

$$z = 5 \pmod{49}$$

$$(4) \quad z = 25 \pmod{49}$$

$$z = 125 \pmod{49}$$

$$= 37$$

$$(5) \quad z = 37^2 \pmod{49}$$

$$= 1369 \pmod{49}$$

$$\begin{array}{r} 389 \\ 343 \\ \hline 46 \end{array}$$

$$\begin{array}{r} 46 \\ 6 \\ \hline 303 \end{array}$$

$$p = 49$$

$$a = 31$$

$$\text{GCD}(31, 49) = 1$$

$$y \equiv 31^{48} \pmod{49}$$

$$\equiv 1 \pmod{49}$$

Miller-Rabin Primality Testing (p)

let p be a prime

$$p-1 = \underbrace{2^k m}_{\text{even}} \quad 2^k m \quad 2 \nmid m$$

$$a^{p-1} \equiv 1 \pmod{p}$$

$$\underbrace{a^{(p-1)/2}}_{\text{square root}} \equiv \pm 1 \pmod{p}$$

$$\underbrace{a^{2^{k-1} m}}_{\text{square root}} \equiv \pm 1 \pmod{p}$$

$$\boxed{(a^m)^{2^{k-1}} \equiv \pm 1 \pmod{p}}$$

→ necessary condⁿ

Miller-Rabin primality Test

1. choose $a \perp p$

2. $b = a^m \pmod{p}$

if $b \equiv 1 \pmod{p}$

return ("p is prime")

else

for $i = 1$ to $k-1$

{ if $b \equiv -1 \pmod{p}$

then return ("p is prime")

else $b \leftarrow b^2 \pmod{p}$

}

return ("p is composite")

}

AKS primality test

(Agarwal - Kayal - Saxena) primality test

it is deterministically correct for any general number.

Features

The maximum running time of the algo can be expressed as a polynomial over the number of digits in the target number.

Based on the following theorem:

An integer n greater than 2 is prime iff the polynomial congruence relation

$$(x+a)^n \equiv (x^n+a) \pmod{n}$$

$$a \perp n$$

$x \rightarrow$ just a symbol.

The AKS test evaluates the equality by making complexity dependent on the size of x .

$$(x+a)^n \equiv (x^n+a) \pmod{x^r-1, n}$$

simple

$$(x+a)^n - (x^n+a) = (x^r-1)g + nf$$

f & g are some poly.

To implement
 $a=1$

$$(x+1)^n - (x^n+1)$$

$$(x+1)^3 - (x^3+1) =$$

$$x^3 + 3x^2 + 3x + 1 - x^3 - 1 = 3x^2 + 3x$$

all are divisible by 3 -

$$(x^2+1+2x)^{(n+1)}$$
$$(x^3 + x^2 + 1 + x + 2x^2 + 2x)$$

Security of Diffie Hellman

publically know - $g, p, g^a \bmod p, g^b \bmod p$.
computationally infeasible to calculate
 $g^{ab} \bmod p$.

$$x = g^a \bmod p$$

known known known

To find a is called
Discrete log prob.
[it is NP hard]

$x = g^a$ - easily know using log

* No efficient algo exists
to find discrete log.

Additional requirements on p and g to make
Diffie Hellman more secure -

- ① $g^x \bmod p$ must not equal to 1, unless
 x is a multiple of $(p-1)$.

$$g^x \bmod p = g^{y(p-1)} \bmod p = \underbrace{(g^{(p-1)} \bmod p)^y}_{\bmod p} = 1$$

Reason],

if $g^x \bmod p = 1$ for small x

if $g^3 \bmod p = 1$

$x=3$

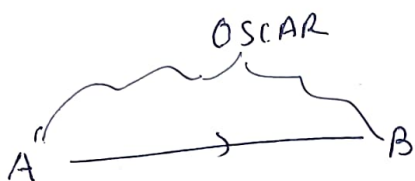
$g^4 \bmod p = g$, $g^5 \bmod p = g^2$, $g^6 \bmod p = 1$

cycle

- ② $\frac{(p-1)}{2}$ must also be a prime
safe prime 3, 7, 11 13 x discrete log. "Pohlig-Hellman Prob"
- if this prop. does not
satisfied, it may be possible
to efficiently compute

Man-in-the-Middle Attack

- Diffie-Hellman algorithm is insecure when there be an intruder ("man-in-the-middle") on the channel b/w ALICE & BOB, who can modify messages sent from ALICE to BOB or vice-versa.

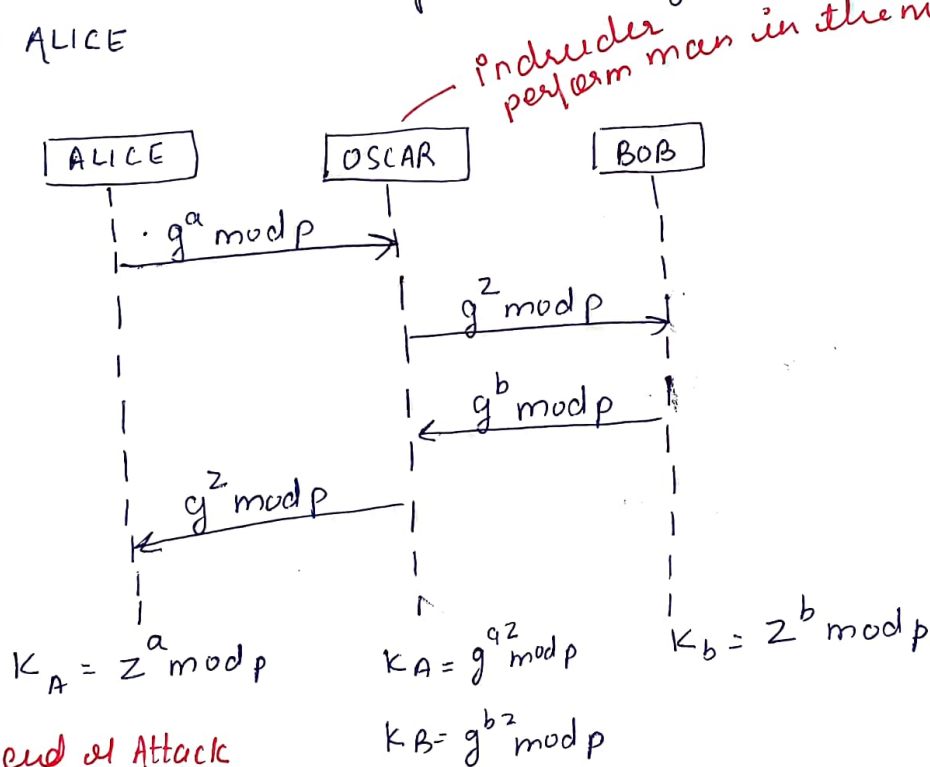


- Suppose g and p publicly known

- when ALICE receives g^a , ALICE cannot found

1. It was really sent by BOB or not

2. It was modified during transit from BOB to ALICE



At the end of Attack

ALICE: $g^{az} \bmod p$ (OSCAR)

BOB: $g^{bz} \bmod p$ (OSCAR)

However, ALICE & BOB think that they have secret shared key with each other.

- After intruder has performed man-in-middle attack:
- when ALICE sends an encrypted message to BOB, intruder can read it and modify it before forwarding it to BOB.
- similarly can read and modify messages from BOB to ALICE
- Only Secure against "passive attacks"

Defences Against Man-in-the-Middle attack

- ① - Suppose after completing the Diffie-Hellman algorithm, ALICE encrypts and transmits the established shared key to Bob to prove that she is indeed ALICE $(g^{ab} \bmod p)$
 will Bob be able to detect man-in-the-middle attack? (NO)
- ② $p, q \rightarrow$ public
 $b \leftarrow$ secret to BOB
 $g^b \bmod p \rightarrow$ public of BOB
 public key of every user is stored in a database like a telephone directory.
 ALICE $\rightarrow$ access database
 BOB $\rightarrow$ It does work (securely connect to the database)
 PKI
 (Public Key Infrastructure)